

ESM Credit LOS — Data Model

Source: backend/prisma/schema.prisma (6,954 lines; **210 models, 104 enums** total). This doc covers the **credit domain only** (~80 models), which spans schema lines **3438–6954** (service-desk / HR / CRM models at lines 18–3437 are excluded except where they are direct FKs of credit models).

Conventions to know:

- Enums (ApplicationState, FacilityType, RiskRating, CommitteeVoteChoice, DisbursementStatus, etc.) are Prisma enums — **single source of truth**. Import from @prisma/client, never re-declare (see warning in credit.types.ts).
- Many entities use **soft delete** (deletedAt); query with deletedAt: null.
- Table names are consistently snake_case (@@map), UUID PKs with @db.Uuid are standard.
- Line numbers are from the schema (they can shift with migrations).

1. God-models (treat with care)

Model	Line	Fields	Relations	Notes
`CreditApplication`	4506	~107	~48	**Central aggregate.** 20-state lifecycle, product, lane, rating, CA-memo fields, LOO, sign-offs; ~40 child relation arrays (facilities, documents, auditEvents, scoreRuns, decisions, recommendations, conditions, memoVersions, committeeAgendaItems, covenants, payments, warningSignals, riskAssessments, deviations, comments, ...)
`BorrowerProfile`	4055	~79	~24	Central borrower aggregate; 1:1 credit extension of `CrmAccount` / `CrmContact`
`User`	79	~163	~110	Platform-wide identity; many named relations back to credit models (Rm/Analyst/MemoLockedBy/etc.)

These aggregate most of the module. Changes here ripple widely.

2. Data-model domains (~80 credit models)

(a) Borrower & parties

Model	Line	Purpose / key FKs
`BorrowerProfile`	4055	Borrower master (type/segment/lifecycle, risk rating, exposure, encrypted PII)
`BorrowerCreditProfile`	4194	Borrower-360 aggregate credit snapshot (score, DSR); `borrowerId` unique

Model	Line	Purpose / key FKs
`BorrowerIncome`	4213	Income & commitment profile; `borrowerId` unique
`BorrowerBureauReport`	4235	Uploaded bureau report (CTOS/CCRIS PDF); children: `BorrowerBureauFacility`
`BorrowerBureauFacility`	4255	Facilities within a bureau report
`BorrowerActivity`	4270	Borrower-360 activity log
`BorrowerRiskRun`	4285	Borrower-level scorecard run (pre-application, immutable)
`FatcaCrsDeclaration`	4322	FATCA/CRS tax declaration
`Director`	4348	Corporate director
`Shareholder`	4380	Corporate shareholder
`UltimateBeneficialOwner`	4410	UBO
`RelatedPartyGroup` / `RelatedPartyMember`	4434/4448	Connected-party/common-ownership group + membership
`ConsentRecord`	3915	PDPA consent per subject/purpose (P1-2)
`BorrowerDuplicateException`	6924	Duplicate-borrower exception register (LOS-017)
`AmlRescreenEvent`	3764	Quarterly AML re-screen log
`SuspiciousTransaction` + `StrAttachment`	3798/3828	STR register (P1-7) + attachments

(b) Application & facilities

Model	Line	Purpose / key FKs
`CreditAppCounter`	3438	Atomic application-number sequence (`CA-...`)
`CreditApplication`	4506	**God-model / central aggregate** (see §1)
`CreditApplicationDraft`	4665	Durable wizard state per user
`BorrowerOnboardingRun`	4675	Idempotent onboarding run
`ApplicationFacility`	4691	Requested facility (amounts, tenor, rate, Islamic variants, SME lane config)
`RequestItem`	4777	CA Memo §3 request items (renewal/variation/policy-breach/SICR)
`PricingWorksheet`	4750	Loan-pricing engine output (§2.1); `facilityId` unique
`ExposureSummary`	4798	Grouped secured/unsecured exposure
`Branch`	4473	Multi-branch reference (also config data)

(c) Collateral & guarantees

Model	Line	Purpose / key FKs
`Collateral`	5216	Security for a facility; dual valuation, haircut inputs; **rich soft-delete** (`softDeletedAt/ById/Reason`)
`CollateralHaircutConfig`	5262	Haircut % by security category (P1-4)
`CollateralApplicationLink`	5276	Cross-application collateral sharing
`CollateralValuation`	5293	Valuation snapshot history
`CollateralLien`	5311	Liens/charges
`InsuranceCover`	5330	Insurance policy on collateral
`Guarantee`	5352	Guarantee on a facility; capacity + related-party checks (P1-5)

(d) Credit assessment / risk / scoring

Model	Line	Purpose
`ExternalRating`	4825	Agency rating
`EclSnapshot` / `EclForecast`	4846/4872	ECL snapshot + Y1/Y2/Y3 forecast
`CashflowProjection` / `ProjectionLineItem`	4893/4905	Cash-flow projection
`SensitivityScenario`	4925	Stress scenario metrics
`FinancialStatement` / `FinancialLineItem` / `FinancialRatio`	5706/5748/5771	Borrower BS/PL/CF statements + line items + ratios
`AccountProfitability` / `ProfitabilityLine`	5385/5400	Profitability report
`WalletShare`	5423	Wallet share per facility type
`KeyCounterparty`	5446	Supplier/buyer/competitor counterparty
`AccountUtilisationSnapshot`	5471	Monthly account utilisation
`QualitativeAssessment`	6022	Qualitative scores (management/relationship/industry/collateral)
`RetailIncome`	6043	Individual borrower DSR assessment (gross + net)
`IndustryAssessment`	6514	Sector outlook
`RiskAssessment`	6556	Risk category assessment + weighted risk engine
`RiskFactorMatrix`	6539	Configurable weighted risk factors
`RmdIssue`	6583	RMD issue log
`EsgAssessment`	6602	ESG assessment
`SicrAssessment`	6621	SICR trigger assessment (corporate committee gate)
`ApplicationSignoff`	6643	Prepared/Reviewed/Concurred sign-offs
`CreditScorecard` / `CreditScorecardVersion`	5798/5916	Scoring model + versioned factor-weights snapshot

Model	Line	Purpose
`ScoreFactorDefinition`	5821	Governed factor definitions, effective-dated + successor chain
`RatingBandConfig`	5884	Score→rating band config, versioned + governed
`CreditScoreRun`	5943	Result of scoring an application (factor scores, rating, overrides, provenance)
`ApplicationAssessmentResult`	5988	**Frozen governed record** populated at committee submission

(e) Committee & approvals

Model	Line	Purpose / key FKs
`CommitteeMeeting` / `CommitteeMember` / `CommitteeAgendaItem` / `CommitteeVote`	6112/6129/6146/6167	Meetings, members/attendance, agenda, votes (quorum)
`CreditDecision`	5501	Approval/rejection decision record; `conditionRecords`→Condition
`CreditRecommendation`	5530	Analyst recommendation; submitted records **immutable + superseded**
`CreditApprovalMatrix` / `CreditApprovalMatrixVersion`	5648/5684	Authority levels by exposure & risk rating + snapshot
`DeviationApproval`	3876	Policy deviation/exception register (P1-6)
`ScoreOverrideApproval`	6339	Dual-approval for score overrides ≥2 notches

(f) Conditions & disbursement

Model	Line	Purpose / key FKs
`Condition`	5569	Precedent/subsequent condition with fulfilment & waiver
`DisbursementOrder`	5612	Disbursement lifecycle control (create→approve→disburse)

(g) Monitoring & early warning

Model	Line	Purpose
`FacilityHealth`	6187	Health status & review cadence
`CovenantDefinition` / `CovenantTest`	6208/6229	Covenant definition + test result (breach → EWS)
`PaymentEvent`	6248	Payment due/paid status (aging)
`EarlyWarningSignal`	6266	EWS with FK-based dedup to covenant/condition

(h) Documents

Model	Line	Purpose / key FKs
`CreditDocument`	4970	Document with classification, sha256, AV scan, verification
`CreditDocumentVersion`	5015	Replacement version tracking
`DocumentRequirement`	5037	Per-application doc checklist

(i) SLA & policy / rules

Model	Line	Purpose / key FKs
`CreditSlaPolicy` / `CreditSlaPolicyBranchOverride` / `CreditSlaBreach`	5155/4492/5189	SLA target state/hours/escalation + branch overrides + breaches
`CreditPolicyLimit`	3852	Exposure/concentration limit
`CreditPolicyParameter`	5060	Generic keyed policy params, scoped by product/lane/borrower
`CreditRuleConfig`	5086	Required-document / required-field rules

(j) Audit & security

Model	Line	Purpose / key FKs
`CreditAuditEvent`	5119	**Append-only audit trail with hash-chain + sequence** (LOS-013)
`PiiReadLog`	6297	PII read-access log
`CreditExportEvent`	6317	CSV/PDF export DLP log
`CreditMemoVersion`	5859	**Immutable versioned memo snapshot**, lockable
`ApplicationComment`	6784	Collaboration thread (P2-4)

(k) Config / reference data

Branch (4473), CreditFxRate (6765, unique currency+effectiveDate), RatingBandConfig, RiskFactorMatrix, CollateralHaircutConfig, SchedulerConfig (6664, credit jobs).

3. Key enums used by credit models (defined in schema)

Enum	Used by
`ApplicationState` (3447)	CreditApplication.state — 19 states (DRAFT...CLOSED/WITHDRAWN/REFERRED_BACK)
`BorrowerType` / `BorrowerSegment` / `BorrowerLifecycleStatus` (3470/3477/3483)	BorrowerProfile
`ProcessingLane` (3490)	CreditApplication.lane (PERSONAL_FAST / SME / CORPORATE)
`RiskRating` (3504)	BorrowerProfile, CreditApplication, BorrowerRiskRun, CreditScoreRun, CreditApprovalMatrix, RatingBandConfig (AAA...D, NR)
`AmlRiskTier` (3519)	BorrowerProfile

Enum	Used by
`ApplicationType` (3528)	CreditApplication (NEW / ADDITIONAL / RENEWAL / VARIATION)
`AccountClassification` (3539)	CreditApplication, SocrAssessment (BNM-aligned, PERFORMING...IMPAIRED)
`CreditProductType` (3554)	CreditApplication, CreditScorecard, CreditPolicyParameter, CreditRuleConfig
`FacilityType` (3566)	ApplicationFacility (incl. Islamic RWC_I/LC_I/BG_I/ICMTD_I/CASHLINE)
`SecurityCategory` (3583)	Collateral, CollateralHaircutConfig
`CounterpartyRole` (3594)	KeyCounterparty
`MfrsStage` (3601)	EclSnapshot, EclForecast
`DocumentClass` (3631)	CreditDocument, DocumentRequirement, CreditRuleConfig
`CovenantType` / `CovenantFrequency` (3662/3994)	CovenantDefinition, FacilityHealth
`ConditionType` / `ConditionCategory` / `ConditionStatus` (3674/3680/3689)	Condition
`ApprovalDecisionType` (3697)	CreditDecision
`RejectionReasonCode` (3707)	CreditApplication
`ConsentPurpose` / `ConsentStatus` (3736/3743)	ConsentRecord
`CommitteeVoteChoice` (6100)	CommitteeVote (APPROVE/REJECT/ABSTAIN)
`EarlyWarningSeverity` (3952)	EarlyWarningSignal
`HealthStatus` (3960)	FacilityHealth
`DisbursementStatus` (3967)	DisbursementOrder
`PaymentStatus` (3974)	PaymentEvent
`SignalType` (3982)	EarlyWarningSignal
`FinancialStatementType` / `FinancialPeriod` / `FinancialStatus` / `RatioCategory` (4016/4023/4029/4036)	FinancialStatement, FinancialRatio
`BaseRateType` (4742)	PricingWorksheet (BLR/OPR/FIXED/SORA/KLIBOR)
`CommitteeMeetingStatus` / `CommitteeMemberRole` / `CommitteeAttendance` (6076/6088/6094)	Committee models
`BureauProvider` (6391)	CreditBureauCheck — **CCRIS is historical-only** (non-bank lender, no live CCRIS)
`RiskCategory` (6405)	RiskAssessment
`EsgGuidingPrinciple` / `EsgCategory` (6413/6421)	EsgAssessment
`SocrTriggerType` (6430)	SocrAssessment
`SignoffRole` (6437)	ApplicationSignoff
`ScoreOverrideStatus` (6374)	ScoreOverrideApproval
`RepaymentType` / `RepaymentFrequency` (6869/6877)	ApplicationFacility
`DuplicateExceptionStatus` (6916)	BorrowerDuplicateException
`CurrencyCode` (4001)	CreditApplication, FinancialStatement

> FacilityType Phase-2 (Islamic) variants are hidden behind a feature flag; Phase-1 types are the default dropdown set.

4. Integrity invariants to preserve

- **Audit chain** (CreditAuditEvent, **5119**): append-only; SHA-256 hash + hashVersion; sequence **is the ordering key (not createdAt, which is ms-precision)** — @@unique([applicationId, sequence]) enforces a single predecessor. FK to application has onDelete: Restrict — an application with audit events cannot be deleted. Never rewrite or delete audit records.
- **CA memo versioning** (CreditMemoVersion, **5859**): immutable snapshots; isLocked/lockedAt/lockedBy — on committee submission the latest version is locked and cannot be regenerated.
- **Frozen assessment** (ApplicationAssessmentResult, **5988**): status FROZEN|SUPERSEDED + version; populated at committee submission; later input changes do **not** mutate it.
- **Immutable recommendations** (CreditRecommendation, **5530**): submitted records are immutable; a newer one supersedes the prior (self-FK RecommendationSuccession).
- **Version snapshots**: CreditApprovalMatrixVersion, CreditScorecardVersion, RatingBandConfig carry JSON snapshots for audit trail.
- **Factor governance** (ScoreFactorDefinition, **5821**): effective-dated succession chain (predecessorId self-FK), @@unique([factorKey, effectiveFrom]).
- **Document evidence retention**: documents backing a decision (from APPROVED/OFFER/ACCEPTED onward) cannot be deleted (enforced in stateGuard.util.ts); corrections are new versions.
- **Soft delete**: deletedAt ON BorrowerProfile, CreditApplication, ApplicationFacility, CreditDocument, FinancialStatement; Collateral uses a richer softDeletedAt/ById/Reason set (P1-4, replaces hard delete for audit). Query with deletedAt: null.
- **PII encryption at rest (AES-256-GCM)**:
BorrowerProfile.annualIncomeEncrypted/netWorthEncrypted/sourceOfWealthEncrypted;
Director/Shareholder/UltimateBeneficialOwner.nricPassportEncrypted + nricPassportHmac;
FatcaCrsDeclaration.usTinEncrypted.
- **Canonical dedup forms (LOS-017)**: BorrowerProfile.registrationNumberNormalized / nricPassportNormalized — maintained by the service, never set by clients.
- **Optimistic concurrency**: CreditApplication.version Int default(1) — incremented on every update/transition; race-safe state writes use updateMany WHERE state guards (409).
- **Denormalised exposure**: BorrowerProfile exposure kept in lockstep with canonical computeBorrowerExposure() (G-09) — transitions into/out of exposure states trigger a refresh.
- **Consent immutability**: ConsentRecord unique on (subjectId, subjectType, purpose, status) — one active consent per purpose.

5. Working with the schema

- Migrations: cd backend && npx prisma migrate dev (local) — never destructive commands in production.
- After schema changes: npm run prisma:generate then regenerate client.
- Seed reference data (scorecards, rating bands, policy limits, SLA policies, rule config) via npm run prisma:seed.

- Design markers preserved in the schema: CA Memo Phases 1–5 map fields to memo sections 1–19; ProcessingLane and SME statement classification drive lane routing and approval depth; BNM alignment notes state Citadel does **not** pull live CCRIS (non-bank lender).